CRICKETLENS | SESSION 02

Real IPL Data Audit & First ML Baseline

Development evidence | 17 October 2026 | IT462 Exploratory Data Analysis

This progress report documents measured results from the user-supplied Cricsheet men's IPL JSON archive. The dashboard and both final ML models are NOT finished. This report is an intermediate development record.

1. Verified input and generated tables

Evidence	Observed result
Source ZIP	1,243 valid match JSONs; 2008-04-18 through 2026-05-31
Competition	All 1,243 records marked male / Indian Premier League
Ordinary innings	2,480 (Super Overs excluded)
Overs / ball events	47,756 overs / 295,557 events including illegal deliveries
Player batting appearances	18,768
Next-over training windows	44,714 eligible completed-over checkpoint examples

2. Screening and cricket-specific cleaning

Normalized genuine franchise renames; did not combine unrelated historical franchises. Wides and no-balls do not count as legal balls; no-balls can count as balls faced. Retired hurt does not count as a team wicket. Total team runs, wickets and legal deliveries reconcile between delivery and inning totals across all 2,480 ordinary innings.

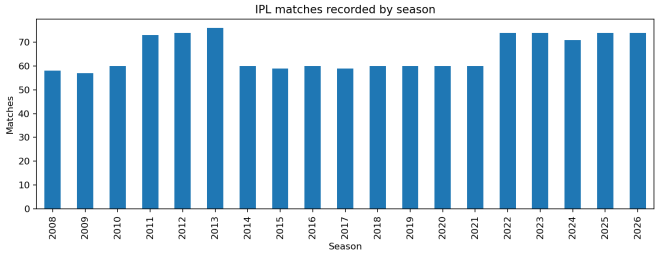
Data-quality observation	Count and treatment
Future next over incomplete (censored)	550: excluded except known early all-out positives
Overs with more than six legal balls	4: flagged, invalid windows excluded
Recorded overs with zero legal balls	4: flagged, not standard full overs
Invalid following-over windows (>6)	4: excluded; some issue categories overlap
City absent	51 matches (4.10%): left missing; no invented cities
Winner empty	25 matches (16 ties, 9 no-results): structurally missing

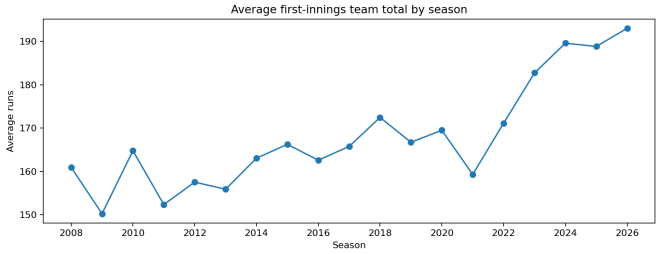
Cleaning caution: 562 screening-issue rows do not mean 562 invalid matches. Absence of D/L outcome method in an ordinary match is also structural, not automatically a defect. Rain and early chase completion can censor next-over outcomes.

3. First descriptive EDA

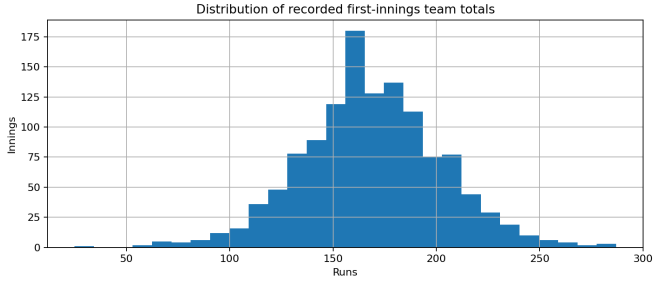
Across the ordinary-innings records: 401,423 total team runs, 14,655 team wickets and 284,465 legal deliveries. The mean first-innings total is 168.57 runs (median 168). Batter milestone rates are 25.53% for 30+ and 11.27% for 50+, conditional on having a recorded batting appearance. The eligible next-over wicket-event rate is 27.70%. These are descriptions, not deployed model predictions.

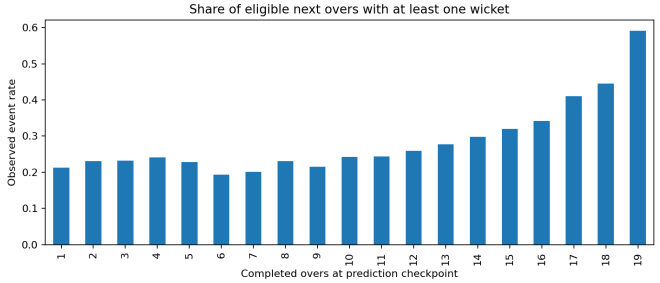
Match counts and scoring trend





Distribution and wicket-event pattern





Interpretation caution: innings may be shortened by match conditions; mean totals across years mix different playing environments. Next-over wicket rates are computed only for eligible non-censored windows and should not be mistaken for unbiased event rates in every possible match state.

4. First model validation (2025 only)

A preliminary logistic-regression model estimates the probability of at least one team wicket in the next over. Features: current runs, wickets, completed overs, run rate, innings, batting team and bowling team. Fit used 39,503 samples through 2024. The 2025 validation set contained 2,596 samples. The 2026 holdout has not been evaluated.

2025 validation metric	Logistic regression	Constant-rate baseline
Brier Score (lower better)	0.19831	0.20319
Log loss (lower better)	0.58561	0.59638
ROC-AUC (higher better)	0.59223	0.50000

Interpretation: the initial model improves only modestly over predicting a constant historical wicket-event probability. We need careful feature engineering, calibration and comparison with more algorithms. These validation results are not a guarantee of success in future IPL games.

5. GitHub-ready deliverables

Location	Contents
src/cricketlens/pipeline.py	JSON extraction, inning/over tables, legal-ball and censoring rules
src/cricketlens/data_audit.py	Season summary, missingness and reconciliation checks, four plots
src/cricketlens/train_wicket_baseline.py	Pre-2025 classifier; year-based validation without 2026 test
docs/DEV_LOG.md	Dated experimental decisions, measured numbers and known limitations
reports/REPORT_DRAFT.md	Living academic EDA report with measured evidence
tests/	Six passing engineering tests

6. Next milestones

Deepen Stage II: inspect venue aliases, truncated/rain games and player identifiers. Stage III: descriptive summaries (SD, IQR, MAD, boxplots), team/venue comparisons, Pearson/Spearman and regression analysis. ML: improve the wicket predictor with time-safe features, then build the pre-match 30+/50+ model. Only after that: FastAPI backend and six-tab React dashboard.

Repository hygiene: keep Cricsheet raw ZIP, generated 35 MB delivery CSV and fitted model files out of GitHub; .gitignore is already configured. Add updated code/docs to local GitHub Desktop checkout, commit and push.

Academic integrity: cite Cricsheet and disclose AI assistance as required by the IT462 course syllabus. This report is a draft to review, not a claim of finished coursework.